Bayesian Model Averaging

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Bayesian model averaging

Let $\{M_\gamma : \gamma \in \Gamma\}$ indicate a set of models for a particular data set y. If Δ is a quantity of interest, e.g. effect size, a future observable, or the utility of a course of action, then its posterior distribution is

$$p(\Delta|y) = \sum_{\gamma \in \Gamma} p(\Delta|M_\gamma, y)p(M_\gamma|y)$$

where

$$p(M_\gamma|y) = \frac{p(y|M_\gamma)p(M_\gamma)}{p(y)} = \frac{p(y|M_\gamma)p(M_\gamma)}{\sum_{\lambda\in\Gamma} p(y|M_\lambda)p(M_\lambda)}$$

is the posterior model probability and

$$p(y|M_\gamma) = \int p(y|\theta_\gamma, M_\gamma)p(\theta_\gamma|M_\gamma)d\theta_\gamma$$

is the marginal likelihood for model M_γ and θ_γ is the set of parameters in model M_γ.

Bayesian model averaged moments

Since $p(\Delta|y)$ is a discrete mixture, we may be interested in simplifying inference concerning Δ to a couple of moments. Let $\hat{\Delta}_\gamma = E[\Delta|y, M_\gamma]$. Then the expectation is

$$E[\Delta|y] = \sum_{\gamma \in \Gamma} \hat{\Delta}_\gamma p(M_\gamma|y)$$

and the variance is

$$Var[\Delta|y] = \left[\sum_{\gamma \in \Gamma} (Var[\Delta|y, M_\gamma] + \hat{\Delta}_\gamma^2) p(M_\gamma|y) \right] - E[\Delta|y]^2$$

The appealing aspect here is that the moments only depend on the moments from each individual model and the posterior model probability.

Difficulties with BMA

- Evaluating the summation can be difficult since $|\Gamma|$, the cardinality of Γ, might be huge.
- Calculating the marginal likelihood.
- Specifying the prior over models.
- Choosing the class of models to average over.

Reducing cardinality

If $|\Gamma|$ is small enough, we can enumerate all models and perform model averaging exactly. But if $|\Gamma|$ is too large, we will need some parsimony.

Rather than summing over Γ, we can only include those models whose posterior probability is sufficiently large

$$\mathcal{A} = \left\{ M_\gamma : \frac{\max_\lambda p(M_\lambda|y)}{p(M_\gamma|y)} = \frac{\max_\lambda p(y|M_\lambda)p(M_\lambda)}{p(y|M_\gamma)p(M_\gamma)} \le C \right\}$$

relative to other models where C is chosen by the researcher. Also, appealing to Occam's razor, we should exclude complex models which receive less support than sub-models of that complex model, i.e.

$$\mathcal{B} = \left\{ M_\gamma : \forall M_\lambda \in \mathcal{A}, M_\lambda \subset M_\gamma, \frac{p(M_\lambda|y)}{p(M_\gamma|y)} < 1 \right\}$$

So, we typically sum over the smaller set of models $\Gamma' = \mathcal{B}$.

Searching through models

One approach is to search through models and keep a list of the best models. To speed up the search the following criteria can be used to decide what models should be kept in Γ':

- When comparing two nested models, if a simpler model is rejected, then all submodels of the simpler model are rejected.
- When comparing two non-nested models, we calculate the ratio of posterior model probabilities

$$\frac{p(M_\gamma|y)}{p(M_{\gamma'}|y)}$$

  if this quantity is less than O_L, we reject M_γ and if it is greater than O_R we reject $M_{\gamma'}$.

Using MCMC to search through models

Construct a neighborhood around $M^{(i)}$ (the current model in the chain), call it $nbh(M^{(i)})$. Now propose a draw M^* from the following proposal distribution

$$q(M^*|M^{(i)}) = \left\{ \begin{array}{cc} 0 & \forall M^* \notin nbh(M^{(i)}) \\ \frac{1}{|nbh(M^{(i)})|} & \forall M^* \in nbh(M^{(i)}) \end{array} \right.$$

Set $M^{(i+1)} = M^*$ with probability $\min\{1, \rho(M^{(i)}, M^*)\}$ where

$$\rho(M^{(i)}, M^*) = \frac{p(M^*|y)}{p(M^{(i)}|y)} \frac{|nbh(M^{(i)})|}{|nbh(M^*)|}$$

and otherwise set $M^{(i+1)} = M^{(i)}$. This Markov chain converges to draws from $p(M_\gamma|y)$ and therefore can estimate posterior model probabilities.

Marginal likelihoods

Some data model and prior combinations have closed form marginal likelihoods, e.g.

$$\begin{aligned} Y|\theta &\sim Bin(n,\theta), \\ \theta &\sim Be(a,b) \qquad\qquad \implies \\ Y &\sim BetaBin(n,a,b) \end{aligned}$$

and

$$\begin{aligned} Y|\mu,\sigma^2 &\sim N_n(\mu 1, \sigma^2 \mathrm{I}), \\ \mu|\sigma^2 &\sim N(m, \sigma^2 v), \\ \sigma^2 &\sim IG(a,b) \qquad\qquad\qquad \implies \\ Y &\sim t_{2a}(m, b[v11^\top + \mathrm{I}]/a), \end{aligned}$$

i.e. a multivariate, location-scale t-distribution.

Unfortunately, the vast majority of marginal likelihoods are not available in closed form.

Estimating the marginal likelihood

Recall the marginal likelihood is

$$\begin{aligned} p(y) &= \int p(y|\theta)p(\theta)d\theta \\ &= E_{p(\theta)}[p(y|\theta)] \\ &\approx \frac{1}{M}\sum_{m=1}^{M} p\left(y|\theta^{(m)}\right), \quad \theta^{(m)} \stackrel{ind}{\sim} p(\theta) \end{aligned}$$

which is a Monte Carlo approach.

Estimating the marginal likelihood

Alternatively,

$$\begin{aligned} p(y) &= \int p(y|\theta) \frac{p(\theta)}{q(\theta)} q(\theta) d\theta \\ &= E_{q(\theta)}\left[\frac{p(\theta)}{q(\theta)} p(y|\theta)\right] \\ &\approx \frac{1}{M}\sum_{m=1}^{M} \frac{p\left(\theta^{(m)}\right)}{q\left(\theta^{(m)}\right)} p\left(y|\theta^{(m)}\right), \quad \theta^{(m)} \stackrel{ind}{\sim} q(\theta) \end{aligned}$$

which is an importance sampling approach. An alternative estimator is

$$p(y) \approx \frac{\sum_{m=1}^{M} \frac{p\left(\theta^{(m)}\right)}{q\left(\theta^{(m)}\right)} p\left(y|\theta^{(m)}\right)}{\sum_{m=1}^{M} \frac{p\left(\theta^{(m)}\right)}{q\left(\theta^{(m)}\right)}}$$

which is biased but generally has lower variance (see Robert and Casella, pg 94).

Evaluating the marginal likelihood

Recall that as the sample size n increases, the posterior converges to a normal distribution (Bernstein von Mises Theorem). Let

$$g(\theta) = \log(p(y|\theta, M)p(\theta|M)) = \log p(y|\theta, M) + \log p(\theta|M)$$

Let $\hat{\theta}_{MAP}$ be the MAP for θ in model M. Taking a Taylor series expansion of $g(\theta)$ around $\hat{\theta}_{MAP}$, we have

$$g(\theta) \approx g(\hat{\theta}_{MAP}) - \frac{1}{2}(\theta - \hat{\theta}_{MAP})A(\theta - \hat{\theta}_{MAP})^\top$$

where A is the negative Hession of $g(\theta)$ evaluated at $\hat{\theta}_{MAP}$. Combining this with the first equation and exponentiating, we have

$$p(y|\theta, M)p(\theta|M) \approx p(y|\hat{\theta}_{MAP}, M)p(\hat{\theta}_{MAP}) \exp\left(-\frac{1}{2}(\theta - \hat{\theta}_{MAP})A(\theta - \hat{\theta}_{MAP})^\top\right)$$

Hence, the approximation to $p(\theta|y, M) \propto p(y|\theta, M)p(\theta|M)$ is normal.

Evaluating the marginal likelihoods (cont.)

If we take the integral over θ of both sides and take the logarithm, we have

$$\log p(y|M) \approx \log p(y|\hat{\theta}_{MAP}, M) + \log p(\hat{\theta}_{MAP}|M) + \frac{p}{2}\log(2\pi) - \frac{1}{2}\log|A|$$

where p is the dimension of θ, i.e. the number of parameters. We call this approximation the Laplace approximation.

Another approximation that is more computationally efficient but less accurate is to only retain terms that increase with n:

- $\log p(y|\hat{\theta}, M)$ increases linearly with n
- $\log |A|$ increases as $p \log n$

As n gets large $\hat{\theta}_{MAP} \to \hat{\theta}_{MLE}$. Taking these two together we have

$$\log p(y|M) \approx \log p(y|\hat{\theta}_{MLE}, M) - \frac{p}{2}\log n$$

Multiplying by -2, we obtain Schwarz's Bayesian Information Criterion (BIC)

$$BIC = -2\log p(y|\hat{\theta}_{MLE}, M) + p\log n$$

Priors over models

For data-based comparisons of models, you can use Bayes Factors directly since

$$BF(M_\gamma : M_{\gamma'}) = \frac{p(y|M_\gamma)}{p(y|M_{\gamma'})} = \frac{\int p(y|\theta_\gamma)p(\theta_\gamma|M_\gamma)d\theta_\gamma}{\int p(y|\theta_{\gamma'})p(\theta_{\gamma'}|M_{\gamma'})d\theta_{\gamma'}}$$

where the last equality is a reminder that priors over parameters still matter.

For model averaging, you need to calculate posterior model probabilities which require specification of the prior probabability of each model. One possible prior for regression models is

$$p(M_\gamma) = \prod_{i=1}^{p} w_i^{\gamma_i}(1-w_i)^{1-\gamma_i}$$

Setting $w_i = 0.5$ corresponds to a uniform prior over the model space.

BMA output

The quantities of interest from BMA are typically

- Posterior model probabilities $p(M_\gamma|y)$
- Posterior inclusions probabilities (for regression)

$$p(\text{including explanatory variable } i|y) = \sum_{\gamma\in\Gamma} p(M_\gamma|y)\mathrm{I}(\gamma_i = 1)$$

  which provides an overall assessment of whether explanatory variable i is important or not.

- Posterior distributions, means, and variances for "parameters", e.g.

$$E[\theta_i|y] = \sum_{\gamma\in\Gamma} E[\theta_{\gamma,i}|y]p(M_\gamma|y)$$

  But does this make any sense? What happened to θ_γ?

- Predictions:

$$p(\tilde{y}|y) = \sum_{\gamma\in\Gamma} p(M_\gamma|y)p(\tilde{y}|M_\gamma, y)$$

R packages for BMA

There are two main packages for Bayesian model average in R

- BMA: glm model averaging using BIC
- BMS: lm model averaging using g-priors and (possibly) MCMC

Back up on CRAN is

- BAS: lm model averaging with a variety of priors and (possibly) MCMC (additionally performed sampling without replacement)

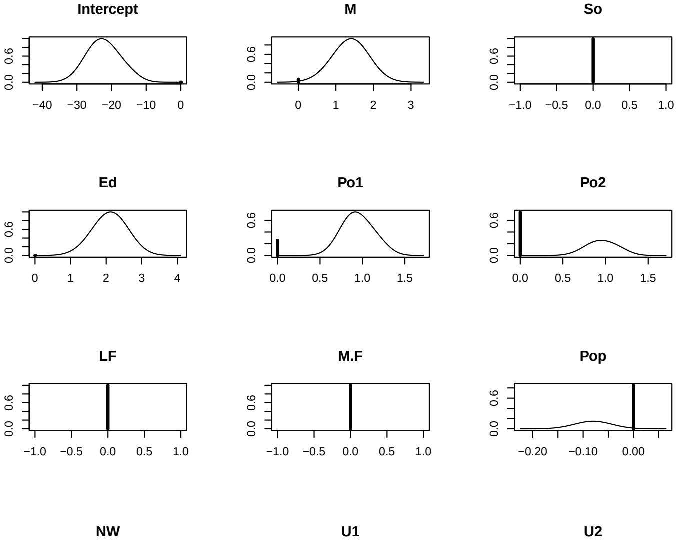
BMA in R

```r
library(BMA)
UScrime <- MASS::UScrime

# Set up data
x = UScrime[,-16]
y = log(UScrime[,16])
x[,-2] = log(x[,-2])

# Run BMA using BIC
lma = BMA::bicreg(x, y,
             strict = TRUE, # remove submodels that are less likely
             OR = 20)       # maximum BF ratio
```

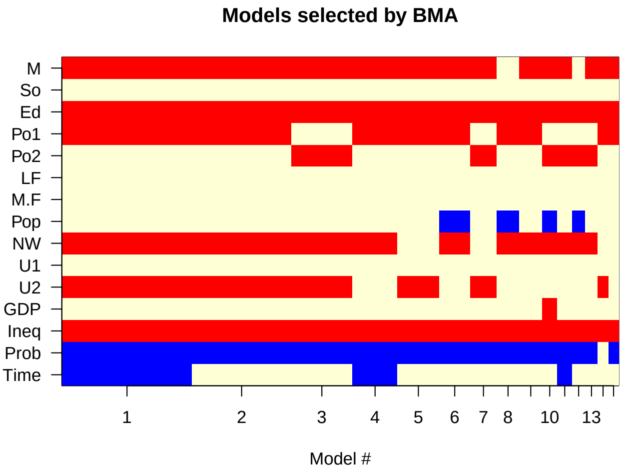
```
summary(lma)

##
## Call:
## BMA::bicreg(x = x, y = y, strict = TRUE, OR = 20)
##
##
##   15  models were selected
##  Best  5  models (cumulative posterior probability =  0.6774 ):
##
##            p!=0    EV         SD        model 1    model 2    model 3    model 4    model 5
## Intercept  100.0  -22.23688  5.10541  -22.63715  -24.38362  -24.50477  -18.26547  -26.29204
## M           93.5    1.29995  0.59146    1.47803    1.51437    1.46061    1.12775    1.68239
## So           0.0    0.00000  0.00000       .          .          .          .          .
## Ed         100.0    2.10530  0.51224    2.22117    2.38935    2.39875    1.83590    2.08662
## Po1         74.2    0.70466  0.44869    0.85244    0.91047       .        0.89054    1.14936
## Po2         25.8    0.24704  0.43075       .          .        0.90689       .          .
## LF           0.0    0.00000  0.00000       .          .          .          .          .
## M.F          0.0    0.00000  0.00000       .          .          .          .          .
## Pop         14.8   -0.01189  0.03188       .          .          .          .          .
## NW          84.1    0.08479  0.05307    0.10888    0.08456    0.08534    0.11670       .
## U1           0.0    0.00000  0.00000       .          .          .          .          .
## U2          66.3    0.20895  0.18420    0.28874    0.32169    0.32977       .        0.33564
## GDP          2.8    0.02025  0.13628       .          .          .          .          .
## Ineq       100.0    1.33551  0.32647    1.23775    1.23088    1.29370    1.24269    1.57576
## Prob        98.1   -0.23801  0.10132   -0.31040   -0.19062   -0.20614   -0.33521   -0.16699
## Time        34.1   -0.10186  0.16713   -0.28659       .          .       -0.33177       .
##
## nVar                                      8          7          7          7          6
## r2                                      0.842      0.826      0.823      0.821      0.805
## BIC                                   -55.91243  -55.36499  -54.40788  -53.79094  -53.62430
## post prob                               0.234      0.178      0.110      0.081      0.074
```

Does this make any sense?

```r
plot(lma)
```



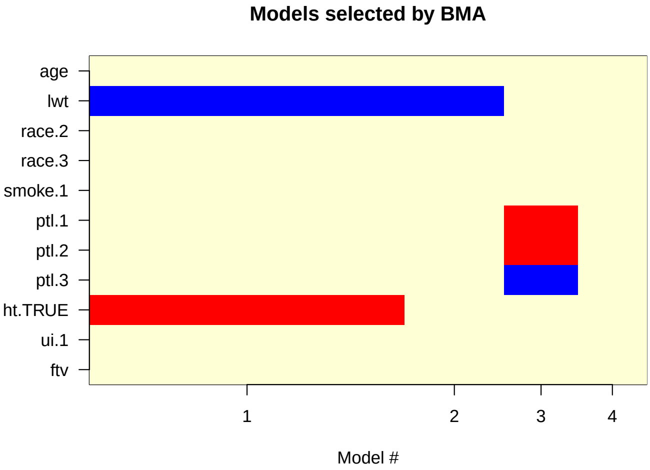
```r
imageplot.bma(lma)
```



BMA for GLMs in R

```r
# Use BIC-based BMA
lma <- BMA::bic.glm(

  x = MASS::birthwt |>
    dplyr::select(-low,-bwt) |>
    mutate(race  = factor(race),     # mother's race (1 = white, 2 = black, 3 = other)
           smoke = factor(smoke),    # smoking status during pregnancy.
           ptl   = factor(ptl),      # number of previous premature labours
           ht    = factor(ht>=1),    # history of hypertension
           ui    = factor(ui)),      # presence of uterine irritability,

  y           = MASS::birthwt$low,   # 1 indicates birthweight < 2.5 kg,
  strict      = TRUE,                # remove models with more likely submodels
  OR          = 20,                  # maximum ratio for excluding models

  glm.family  = "binomial",
  factor.type = TRUE)                # include/exclude factor levels as a group
```

```
##
## Call:
## bic.glm.data.frame(x = mutate(dplyr::select(MASS::birthwt, -low,     -bwt), race = factor(race), smoke = fac
##
##
##   4  models were selected
##  Best  4  models (cumulative posterior probability =  1 ):
##
##             p!=0   EV        SD          model 1     model 2     model 3     model 4
## Intercept  100     0.76092  1.230e+00    1.45068     0.99831    -1.05711    -0.79000
## age          0.0   0.00000  0.000e+00       .           .           .           .
## lwt         74.4  -0.01306  9.639e-03   -0.01865    -0.01406       .           .
## race         0.0
##     .2             0.00000  0.000e+00       .           .           .           .
##     .3             0.00000  0.000e+00       .           .           .           .
## smoke        0.0
##      .1            0.00000  0.000e+00       .           .           .           .
## ptl         13.3
##    .1              0.23225  6.179e-01       .           .        1.75026        .
##    .2              0.08647  4.047e-01       .           .        0.65165        .
##    .3             -1.79255  3.216e+02       .           .      -13.50896        .
## ht          56.6
##   .TRUE            1.04938  1.060e+00    1.85551        .           .           .
## ui           0.0
##   .1               0.00000  0.000e+00       .           .           .           .
## ftv          0.0   0.00000  0.000e+00       .           .           .           .
##
## nVar                                       2           1           1           0
## BIC                                   -753.82285  -751.51602  -750.92334  -750.77644
## post prob                                 0.566       0.178       0.133       0.123
```


Models selected by BMA
age
lwt
race.2
race.3
smoke.1
ptl.1
ptl.2
ptl.3
ht.TRUE
ui.1
ftv
1
2
3
4
Model #

Predictions

```
npkBMA = bicreg(x = npk[, c("block","N","P","K")], y = npk$yield)
summary(npkBMA)

##
## Call:
## bicreg(x = npk[, c("block", "N", "P", "K")], y = npk$yield)
##
##
##   31  models were selected
##  Best  5  models (cumulative posterior probability =  0.4714 ):
##
##            p!=0    EV       SD      model 1  model 2  model 3  model 4  model 5
## Intercept  100.0  53.5651  2.6232   55.125   50.742   54.371   56.333   55.717
## block2      47.0   2.0923  2.9592     .       5.892    2.263     .        .
## block3      88.6   5.9034  3.4860    4.833    9.217    5.588     .       4.833
## block4      67.0  -3.5910  3.3608   -5.817     .      -5.062   -7.025   -5.817
## block5      60.9  -3.1264  3.2565   -5.417     .      -4.663   -6.625   -5.417
## block6      34.4   1.2048  2.4195     .       4.792     .        .        .
## N1         100.0   5.6167  1.6870    5.617    5.617    5.617    5.617    5.617
## P1          15.6  -0.1845  0.7817     .        .        .        .      -1.183
## K1          91.4  -3.6402  1.9466   -3.983   -3.983   -3.983   -3.983   -3.983
##
## nVar                                   5        5        6        4        6
## r2                                   0.688    0.674    0.704    0.608    0.698
## BIC                                -12.097  -11.034  -10.150   -9.792   -9.669
## post prob                            0.183    0.107    0.069    0.058    0.054
```

Predictions

```
p = predict(npkBMA, newdata = npk, quantiles = c(.5,.05,.95))
bind_cols(npk, as.data.frame(p$quantiles)) |> head(20)
```

```
##    block N P K yield      0.5     0.05     0.95
## 1      1 0 1 1  49.5 49.74860 40.96947 58.48361
## 2      1 1 1 0  62.8 59.01439 50.23957 67.69668
## 3      1 0 0 0  46.8 53.58095 44.77569 62.30043
## 4      1 1 0 1  57.0 55.54910 46.77694 64.28062
## 5      2 1 0 0  59.8 61.29444 53.15264 69.32592
## 6      2 1 1 1  58.5 57.45633 49.44848 65.42687
## 7      2 0 0 1  55.5 52.02752 44.00540 59.99400
## 8      2 0 1 0  56.0 55.48986 47.40503 63.48321
## 9      3 0 1 0  62.8 57.00405 47.21705 66.71617
## 10     3 1 1 1  55.8 58.97445 49.33077 68.57627
## 11     3 1 0 0  69.5 62.80984 52.96439 72.55522
## 12     3 0 0 1  55.0 53.54594 43.88129 63.14868
## 13     4 1 0 0  62.0 58.34556 48.84541 67.82723
## 14     4 1 1 1  48.8 54.52696 44.85351 64.14974
## 15     4 0 0 1  45.5 49.09411 39.42632 58.71447
## 16     4 0 1 0  44.2 52.54720 43.07168 61.98942
## 17     5 1 1 0  52.0 58.31376 49.07504 67.54200
## 18     5 0 0 0  51.5 52.88006 43.60689 62.15077
## 19     5 1 0 1  49.8 54.85841 45.45391 64.25045
## 20     5 0 1 1  48.8 49.05686 39.65504 58.44869
```

MCMC for sampling θ and M

Suppose, you construct a Markov chain to sample jointly from $p(M_\gamma, \theta_\gamma|y)$. An issue here is that when you move $M_\gamma \to M_{\gamma'}$, there is a chance that you change the dimension of θ, i.e. $\sum_{i=1}^p \gamma_i \neq \sum_{i=1}^p \gamma_i'$. This can be done via Metropolis-Hastings where the change of dimension is taken into account and this approach is called reversible jump MCMC.

An alternative is to fully incorporate γ as a parameter in the model. For example,

$$\begin{array}{rcl} y_{ij} & \stackrel{ind}{\sim} & N(\gamma_i\theta_i, \sigma^2) \\ \theta_i & \stackrel{ind}{\sim} & N(\mu, \tau) \\ \gamma_i & \stackrel{ind}{\sim} & Ber(w_i) \end{array}$$

This is essentially a way to implement the point-mass prior.

MCMC for Model averaging for GLMs

We can implement a similar MCMC to perform model averaging in Bayesian GLMs. Let $\theta_i = E[y_i|\theta_i]$and ϕ as a dispersion parameter, then we can define a GLM as

$$\begin{array}{ll} y_i & \sim p(y_i|\theta_i, \psi) \\ \theta_i & = g^{-1}(X_i'\beta) \\ \beta_j & = \gamma_j \phi_j \\ \phi_j & \stackrel{ind}{\sim} N(\mu, \tau) \\ \gamma_j & \stackrel{ind}{\sim} Ber(w_j) \end{array}$$

For probit and ordinal regression, we can augment the model with parameters ζ_i, e.g. for probit regression

$$y_i = \mathrm{I}(\zeta_i > 0) \text{ and } \zeta_i \stackrel{ind}{\sim} N(\theta_i, \psi).$$

There is a similar augmentation for logistic regression, see the `BayesLogit` R package and references therein. For these models and linear regression, we can construct an MCMC entirely using Gibbs steps. For other models, e.g. Poisson regression, sampling ϕ_j results in a non-Gibbs step.

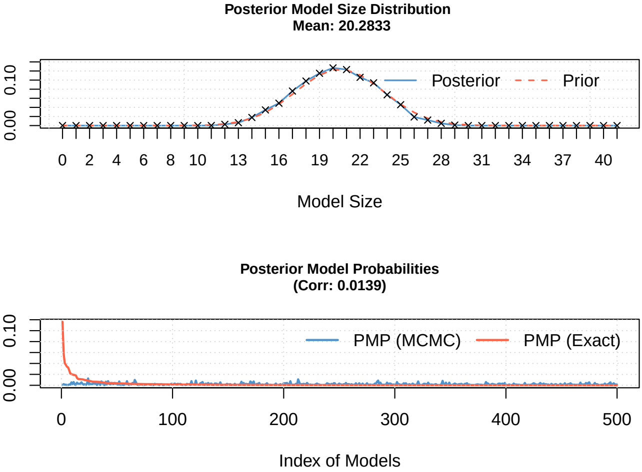
BMS

```r
library(BMS) # Bayesian model sampling
data(datafls)
dim(datafls)

## [1] 72 42

bma1 = bms(
  X.data   = datafls,
  burn     = 1000,
  iter     = 10000,
  g        = "EBL",         # Model specific empirical Bayes estimate
  mprior   = "uniform",     # model over priors (extremely flexible)
  user.int = interactive())
```

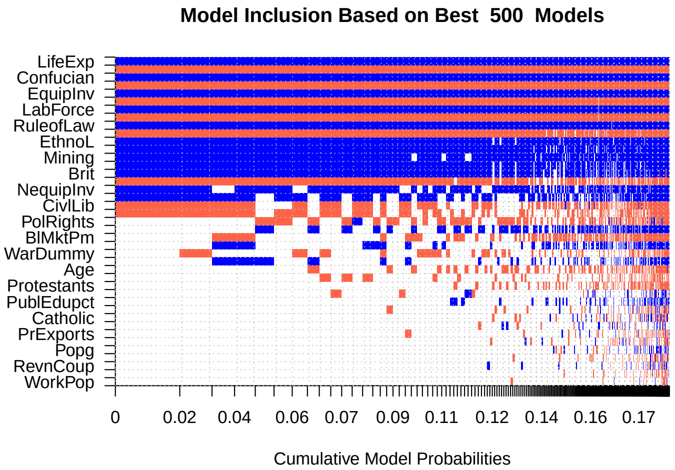
```
print(bma1)

##                PIP      Post Mean      Post SD Cond.Pos.Sign Idx
## GDP60       1.0000 -1.650621e-02 2.953045e-03    0.00000000  12
## Confucian   1.0000  6.131941e-02 1.398614e-02    1.00000000  19
## LifeExp     0.9996  8.888364e-04 2.575080e-04    1.00000000  11
## SubSahara   0.8981 -1.648193e-02 8.208535e-03    0.00000000   7
## EquipInv    0.8833  1.195939e-01 6.293920e-02    1.00000000  38
## Mining      0.8267  3.159094e-02 2.003432e-02    1.00000000  13
## Hindu       0.7548 -5.381439e-02 4.293733e-02    0.01152623  21
## NequipInv   0.7484  3.788405e-02 2.919681e-02    1.00000000  39
## RuleofLaw   0.7326  8.392129e-03 6.806351e-03    1.00000000  26
## LabForce    0.7150  1.797671e-07 1.543203e-07    0.97860140  29
## EcoOrg      0.6804  1.363371e-03 1.215124e-03    1.00000000  14
## LatAmerica  0.6325 -7.014337e-03 7.349975e-03    0.04695652   6
## EthnoL      0.6065  6.679661e-03 6.897232e-03    0.99719703  20
## Spanish     0.5879  5.437040e-03 6.764977e-03    0.96598061   2
## BlMktPm     0.5874 -4.158779e-03 4.450378e-03    0.00000000  41
## Muslim      0.5867  7.309325e-03 8.415066e-03    0.99335265  23
## HighEnroll  0.5742 -5.035253e-02 5.438604e-02    0.00766284  30
## French      0.4724  3.743428e-03 5.173097e-03    0.99153260   3
## English     0.4572 -3.335110e-03 4.885083e-03    0.00000000  35
## Buddha      0.4541  4.809012e-03 7.098296e-03    0.99845849  17
## Protestants 0.4539 -4.472525e-03 6.147630e-03    0.00110156  25
## PrScEnroll  0.4531  7.617071e-03 1.080200e-02    0.97550210  10
## CivlLib     0.4417 -8.883440e-04 1.468537e-03    0.06135386  34
## PolRights   0.4238 -5.847481e-04 1.162588e-03    0.10594620  33
## Age         0.3822 -1.407141e-05 2.527983e-05    0.00366300  16
## Brit        0.3728  1.764676e-03 3.798127e-03    0.81115880   4
## OutwarOr    0.3444 -9.492090e-04 1.859284e-03    0.02642276   8
## Catholic    0.3219  1.520849e-04 4.232549e-03    0.51599876  18
## PrExports   0.3175 -1.788397e-03 4.541944e-03    0.09952756  24
## YrsOpen     0.2986  2.067048e-03 4.650562e-03    0.90790355  15
## WarDummy    0.2869 -8.130271e-04 1.859290e-03    0.00906239   5
## PublEdupct  0.2455  2.550901e-02 8.405159e-02    0.85213849  31
```

```
summary(bma1)

## Mean no. regressors               Draws             Burnins                Time  No. models visited
##           "20.2833"             "10000"              "1000"      "0.27246 secs"              "4337"
##      Modelspace 2^K           % visited         % Topmodels            Corr PMP            No. Obs.
##           "2.2e+12"             "2e-07"                "18"            "0.0139"                "72"
##         Model Prior             g-Prior     Shrinkage-Stats
##    "uniform / 20.5"               "EBL"          "Av=0.961"
```

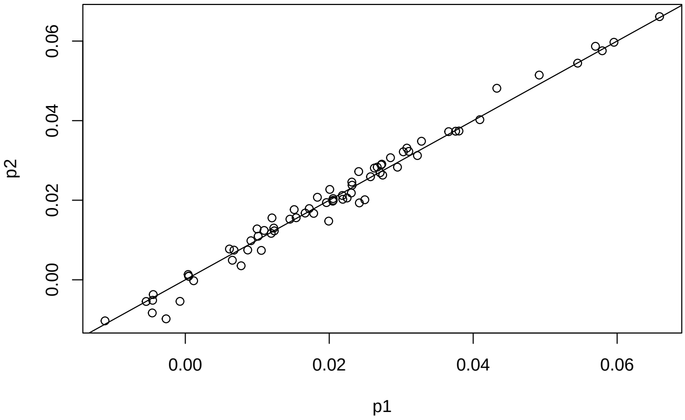
```r
plot(bma1)
```


Posterior Model Size Distribution
Mean: 20.2833
0.00
0.10
Posterior
Prior
0 2 4 6 8 10 13 16 19 22 25 28 31 34 37 40
Model Size
Posterior Model Probabilities
(Corr: 0.0139)
0.00
0.10
PMP (MCMC)
PMP (Exact)
0 100 200 300 400 500
Index of Models

```r
image(bma1)
```


Model Inclusion Based on Best 500 Models
LifeExp
Confucian
EquipInv
LabForce
RuleofLaw
EthnoL
Mining
Brit
NequipInv
CivlLib
PolRights
BlMktPm
WarDummy
Age
Protestants
PublEdupct
Catholic
PrExports
Popg
RevnCoup
WorkPop
0
0.02
0.04
0.06
0.07
0.09
0.11
0.12
0.14
0.16
0.17
Cumulative Model Probabilities

```
p1 = predict(bma1)               # fitted values based on MCMC frequencies
p2 = predict(bma1, exact=TRUE)   # fitted values based on best models
plot(p1,p2); abline(0,1)
```



Model averaging of regression coefficients?

```
set.seed(20171007)
n <- 100
x1 <- rnorm(n)
y <- x1 + rnorm(n)
x2 <- x1 + rnorm(n,0,.01)

library(BMA)
m <- bicreg(cbind(x1,x2), y)

summary(m)
```

```
##
## Call:
## bicreg(x = cbind(x1, x2), y = y)
##
##
##   3  models were selected
##  Best  3  models (cumulative posterior probability =  1 ):
##
##            p!=0    EV       SD       model 1    model 2    model 3
## Intercept  100.0   0.09687  0.08774    0.09686    0.09646    0.09980
## x1          51.6  -0.12069  3.07643      .         0.78988   -7.23838
## x2          54.9   0.91107  3.07714    0.79065      .         8.03022
##
## nVar                                     1          1          2
## r2                                     0.480      0.479      0.483
## BIC                                  -60.81432  -60.67277  -56.82626
## post prob                              0.484      0.451      0.066
```